

# Sumedh Kothari

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## EDUCATION

### Georgia Institute of Technology

Expected May 2028

B.S. in College of Engineering

- **Coursework:** Computer Organization and Programming; Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Ordinary Differential Equations
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout

## EXPERIENCE

### Georgia Tech Renewable Bioproducts Institute | Tong Laboratory

Atlanta, GA

Research Intern

August 2025 – Present

- Used Python, NumPy/Pandas libraries to apply Bayesian statistical modeling to optimize lignin valorization reaction
- Utilized NMR, SEM, and HPLC modeling data to characterize value-added chemicals from lignin breakdown
- Completed comprehensive literature reviews on aromatic compound synthesis, 3D printing custom GO structures, and optimal catalyst design while collaborating with research team to write manuscript for publication

### Hydrogen Engineering Research Consortium

Los Angeles, CA

Simulation Intern

August 2024 – February 2025

- Used MATLAB, dimensionless differential equations to model species' concentrations in eutrophic freshwater ecosystem
- Employed sustainability over sets calculation to determine equilibria, stability of environment; co-authored research paper on environmental systems modeling under UCLA faculty mentorship

## PROJECTS

### IEEE Robotech 2026 Hackathon - Autonomous Track | Georgia Tech

Atlanta, GA

Mechanical/Software Engineer

January 2026

- Designed and CAD-modeled a fully autonomous debris-clearing robot by engineering the chassis, actuation mechanisms, and beacon-deployment system, enabling end-to-end autonomous operation without human intervention
- Trained, developed, and deployed a YOLO-based computer vision model in PyTorch by integrating it into the onboard perception pipeline, achieving 95% debris detection accuracy and enabling real-time hazard identifications
- Integrated perception, navigation, and mechanical actuation systems by linking CV outputs to motion planning and debris-removal hardware, allowing the robot to autonomously clear obstacles to certify landing safety

### Sports Betting Arbitrage Calculator | Python, REST APIs

Atlanta, GA

Developer

November 2025

- Developed application that aggregates live odds from multiple sportsbooks via API, normalizes data, and identifies risk-free arbitrage opportunities across multiple market types (H2H, Spread, Total) with up to 50% daily ROI.
- Built a robust, production-ready pipeline with Dockerization, session management, error handling, and logging, enabling scalable, real-time analysis across dozens of active sports markets.

### AiApply - AI Job Applier | Python, Selenium WebDriver, HTML, JavaScript

Atlanta, GA

Developer

October 2025

- Developed full-stack AI-powered internship application platform to automate end-to-end job search workflows, including resume optimization, cover letter generation, and programmatic application submission on Handshake
- Implemented Selenium-based automation to extract recruiter profiles from filtered company searches and deliver personalized direct messages at scale, increasing outreach efficiency and response rates
- Engineered web scraping algorithms to identify target companies based on location and industry parameters, then automated tailored email outreach highlighting candidate skillsets parsed from resume data

### GreenGrade - Gemini API Developer Competition | Flutter, C++

Los Angeles, CA

Project Lead

July 2024 - August 2025

- Directed 20-person UCLA team to architect and implement a fuzzy logic-driven Sustainability Interval Index calculator with scalable data integration pipelines
- Developed and deployed *GreenGrade*, embedding SII computational model and automating sustainability analytics for 175 mining firms through Gemini API

## SKILLS

- **Technical:** Python, Java, Data analysis (Excel, MATLAB), CAD (SolidWorks)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)